Chapter 3

Planning and Managing the Project

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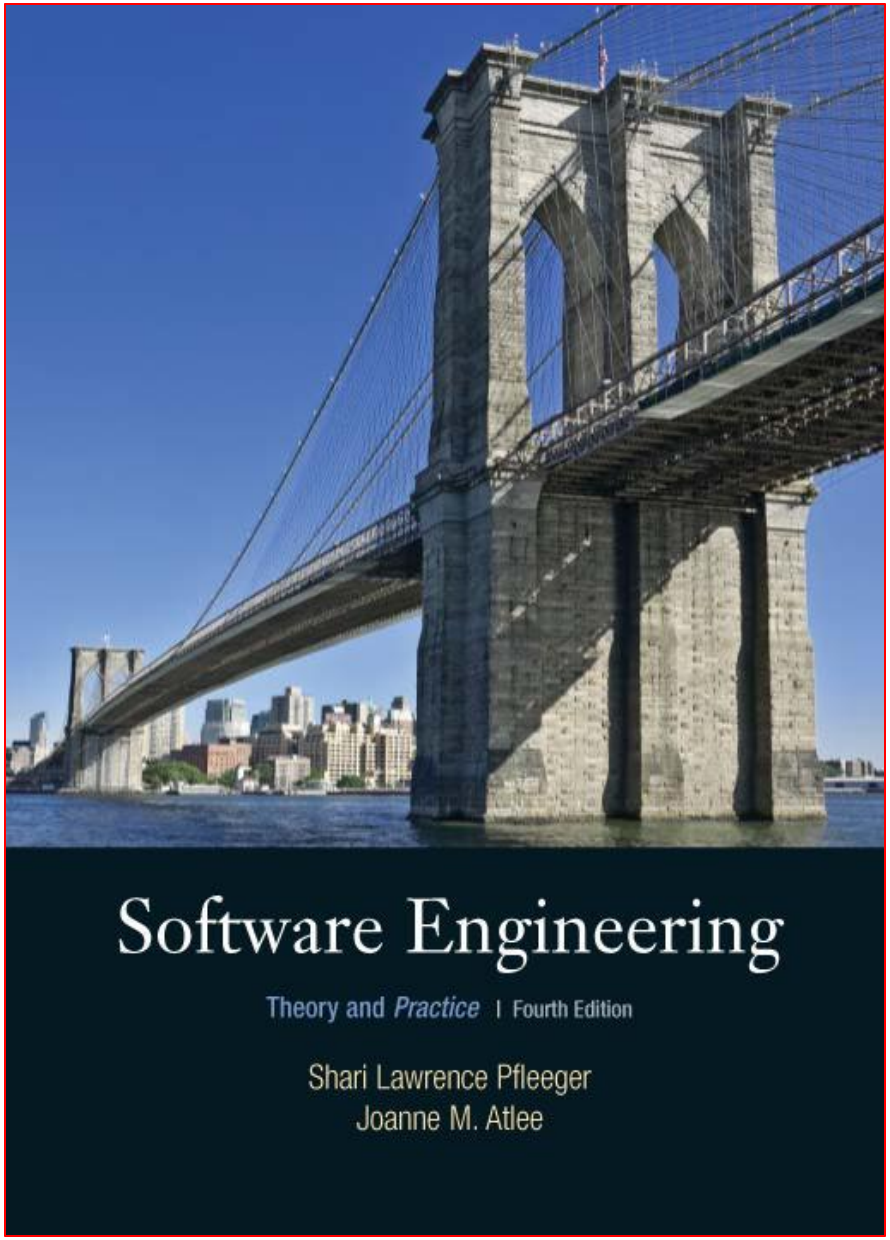


3.1 Tracking Progress

- Do we understand customer's needs?
- Can we design a system to solve customer's problems or satisfy customer's needs?
- How long will it take to develop the system?
- How much will it cost to develop the system?

3.1 Tracking Progress
Project Schedule

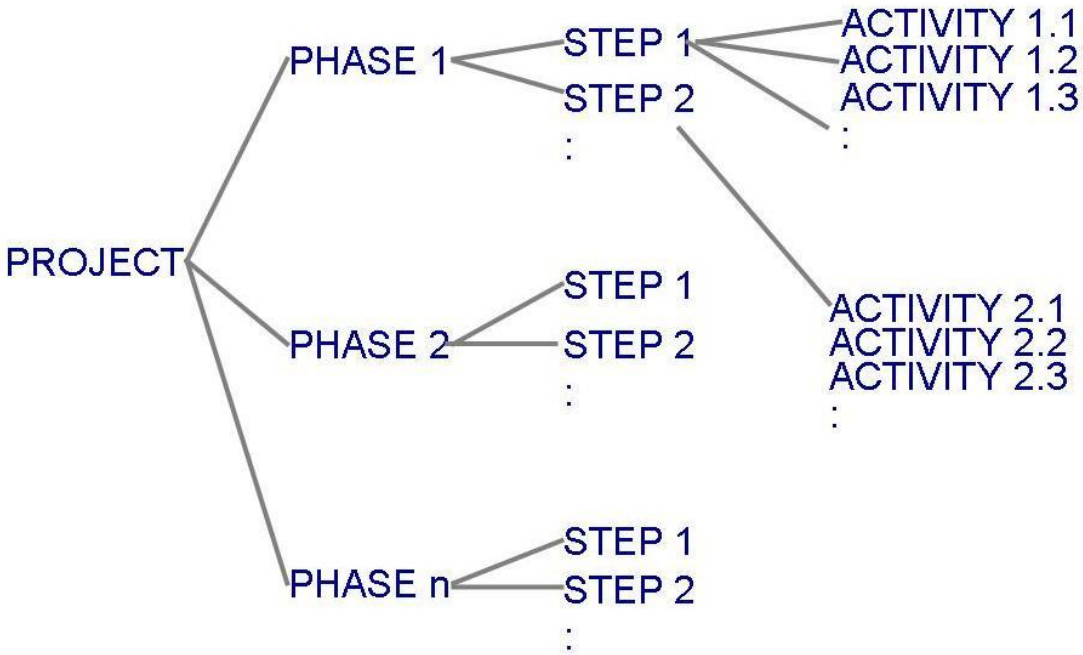
- Describes the software-development cycle for a particular project by
  - enumerating the phases or stages of the project
  - breaking each phase into discrete tasks or activities to be completed
- Portrays the interactions among the activities and estimates the times that each task or activity will take

3.1 Tracking Progress
Milestones and activities

- **Activity**: takes place over a period of time
- **Milestone**: completion of an activity -- a particular point in time
- **Precursor**: event or set of events that must occur in order for an activity to start
- **Duration**: length of time needed to complete an activity
- **Due date**: date by which an activity must be completed

3.1 Tracking Progress
Project Schedule (continued)

- Project development can be separated into a succession of phases which are composed of steps, which are composed of activities



3.1 Tracking Progress
Project Schedule (continued)

- Table 3.1 shows the phases, steps and activities to build a house
  - landscaping phase
  - building the house phase
- Table 3.2 lists milestones for building the house phase

3.1 Tracking Progress

Phases, Steps, and Activities in Building a House

Phase 1: Landscaping the lot	Phase 2: Building the house
Step 1.1: Clearing and grubbing	*Step 2.1: Prepare the site*
Activity 1.1.1: Remove trees	Activity 2.1.1: Survey the land
Activity 1.1.2: Remove stumps	Activity 2.1.2: Request permits
Step 1.2: Seeding the turf	Activity 2.1.3: Excavate for the foundation
Activity 1.2.1: Aerate the soil	Activity 2.1.4: Buy materials
Activity 1.2.2: Disperse the seeds	*Step 2.2: Building the exterior*
Activity 1.2.3: Water and weed	Activity 2.2.1: Lay the foundation
Step 1.3: Planting shrubs and trees	Activity 2.2.2: Build the outside walls
Activity 1.3.1: Obtain shrubs and trees	Activity 2.2.3: Install exterior plumbing
Activity 1.3.2: Dig holes	Activity 2.2.4: Exterior electrical work
Activity 1.3.3: Plant shrubs and trees	Activity 2.2.5: Exterior siding
Activity 1.3.4: Anchor the trees and mulch around them	Activity 2.2.6: Paint the exterior
	Activity 2.2.7: Install doors and fixtures
	Activity 2.2.8: Install roof

3.1 Tracking Progress
Milestones in Building a House

1.1.	Survey complete
1.2.	Permits issued
1.3.	Excavation complete
1.4.	Materials on hand
2.1.	Foundation laid
2.2.	Outside walls complete
2.3.	Exterior plumbing complete
2.4.	Exterior electrical work complete
2.5.	Exterior siding complete
2.6.	Exterior painting complete
2.7.	Doors and fixtures mounted
2.8.	Roof complete
3.1.	Interior plumbing complete
3.2.	Interior electrical work complete
3.3.	Wallboard in place
3.4.	Interior painting complete
3.5.	Floor covering laid
3.6.	Doors and fixtures mounted

Purpose of Creating WBS

- Improve accuracy of cost, time, and resource estimates.
- Define a baseline for performance measurement and control.
- Facilitate clear responsibility assignments.

WBS for Concept Development Project

Task definition: Task 1.1 Concept Scoping

1.1.1 Identify need, benefits and potential customers;
1.1.2 Define desired output/control and input events that drive the application;
Begin Task 1.1.2
1.1.2.1 TR: Review written description of need[14]
1.1.2.2 Derive a list of customer visible outputs/inputs
1.1.2.3 TR: Review outputs/inputs with customer and revise as required;
endtask Task 1.1.2
1.1.3 Define the functionality/behavior for each major function;
Begin Task 1.1.3
1.1.3.1 TR: Review output and input data objects derived in task 1.1.2;
1.1.3.2 Derive a model of functions/behaviors;
1.1.3.3 TR: Review functions/behaviors with customer and revise as required;
endtask Task 1.1.3
1.1.4 Isolate those elements of the technology to be implemented in software;
1.1.5 Research availability of existing software;
1.1.6 Define technical feasibility;
1.1.7 Make quick estimate of size;
1.1.8 Create a Scope Definition;
endtask definition: Task 1.1

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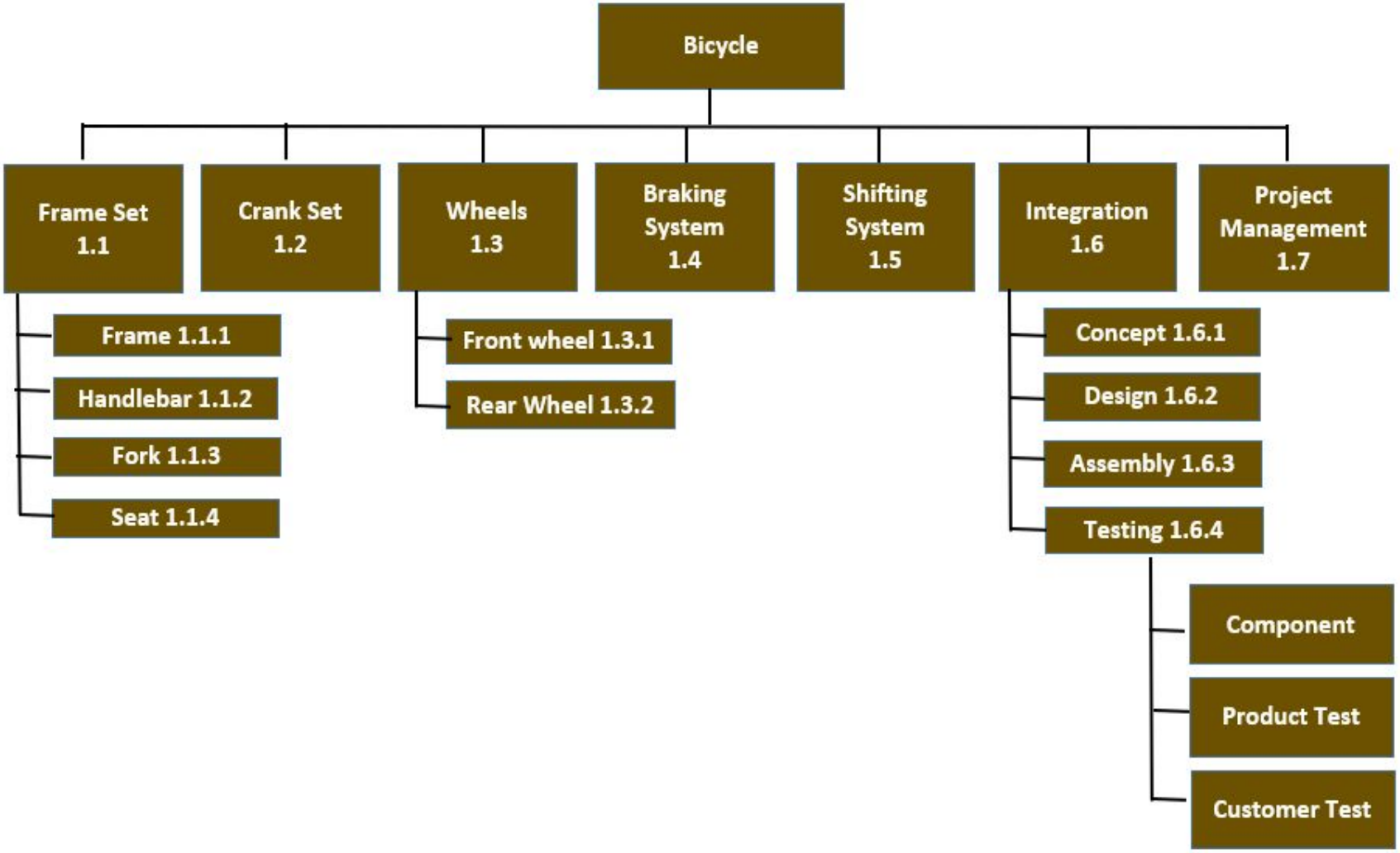
WBS Example


Bicycle
Frame Set 1.1
Crank Set 1.2
Wheels 1.3
Braking System 1.4
Shifting System 1.5
Integration 1.6
Project Management 1.7
Frame 1.1.1
Handlebar 1.1.2
Fork 1.1.3
Seat 1.1.4
Front wheel 1.3.1
Rear Wheel 1.3.2
Concept 1.6.1
Design 1.6.2
Assembly 1.6.3
Testing 1.6.4
Component
Product Test
Customer Test

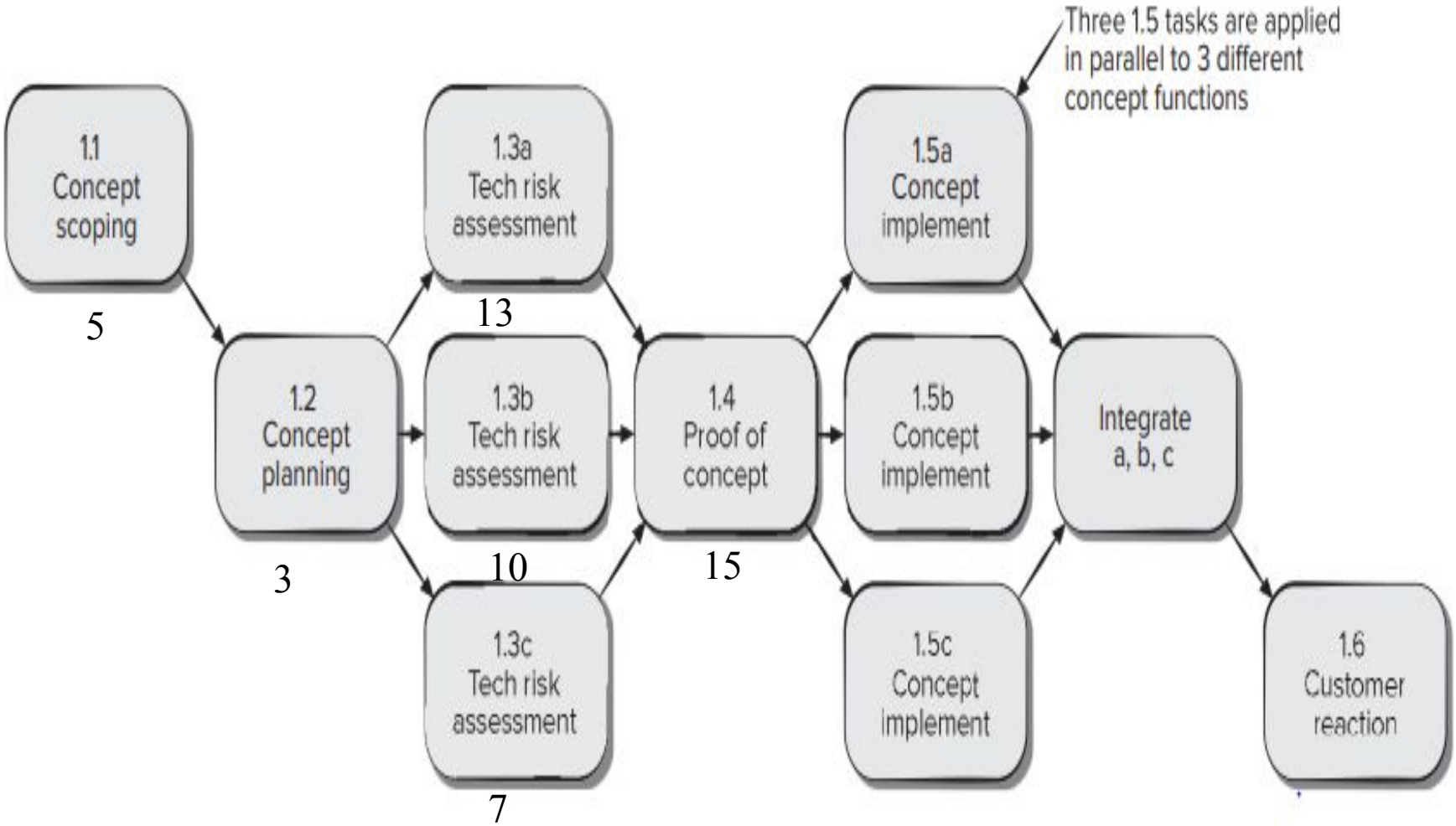
3.1 Tracking Progress
Work Breakdown and Activity Graphs

- Work breakdown structure depicts the project as a set of discrete pieces of work
- Activity graphs (PERT charts) depict the dependencies among activities
  - *Nodes*: project milestones
  - *Lines*: activities involved

Activity networks can be of two types

AoA: Activity on Arc
AoN: Activity on Node

A Task Network for concept development



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3.1 Tracking Progress

Critical Path Method (CPM)

- Minimum amount of time it will take to complete a project
  - Reveals those activities that are most critical to completing the project on time
- **Real time (actual time):** estimated amount of time required for the activity to be completed
- **Available time:** amount of time available in the schedule for the activity's completion
- **Slack time:** the difference between the available time and the real time for that activity

3.1 Tracking Progress

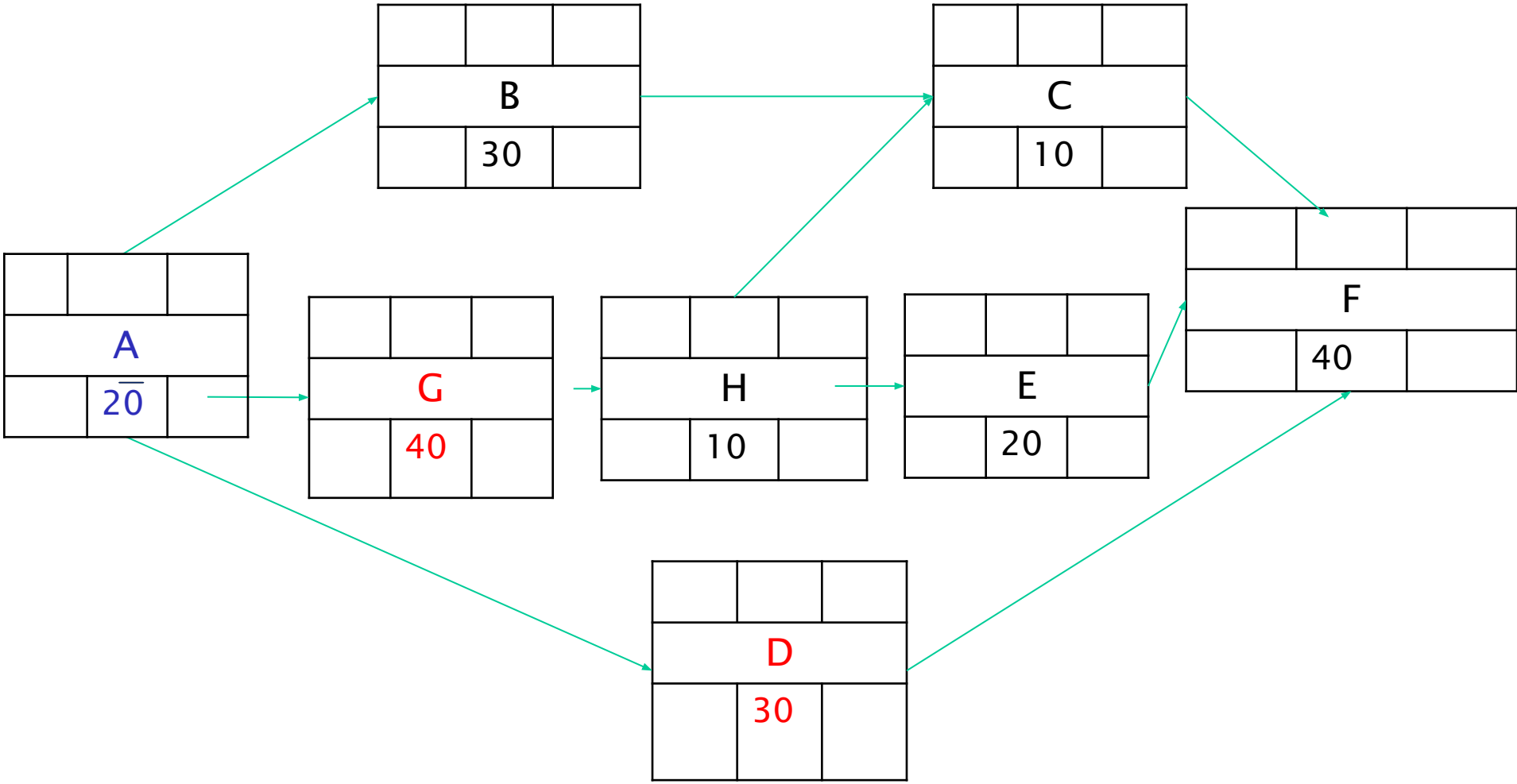
Critical Path Method (CPM) (continued)

- **Critical path**: the slack at every node is zero
  - can be more than one in a project schedule
- **Slack time** = available time – real time

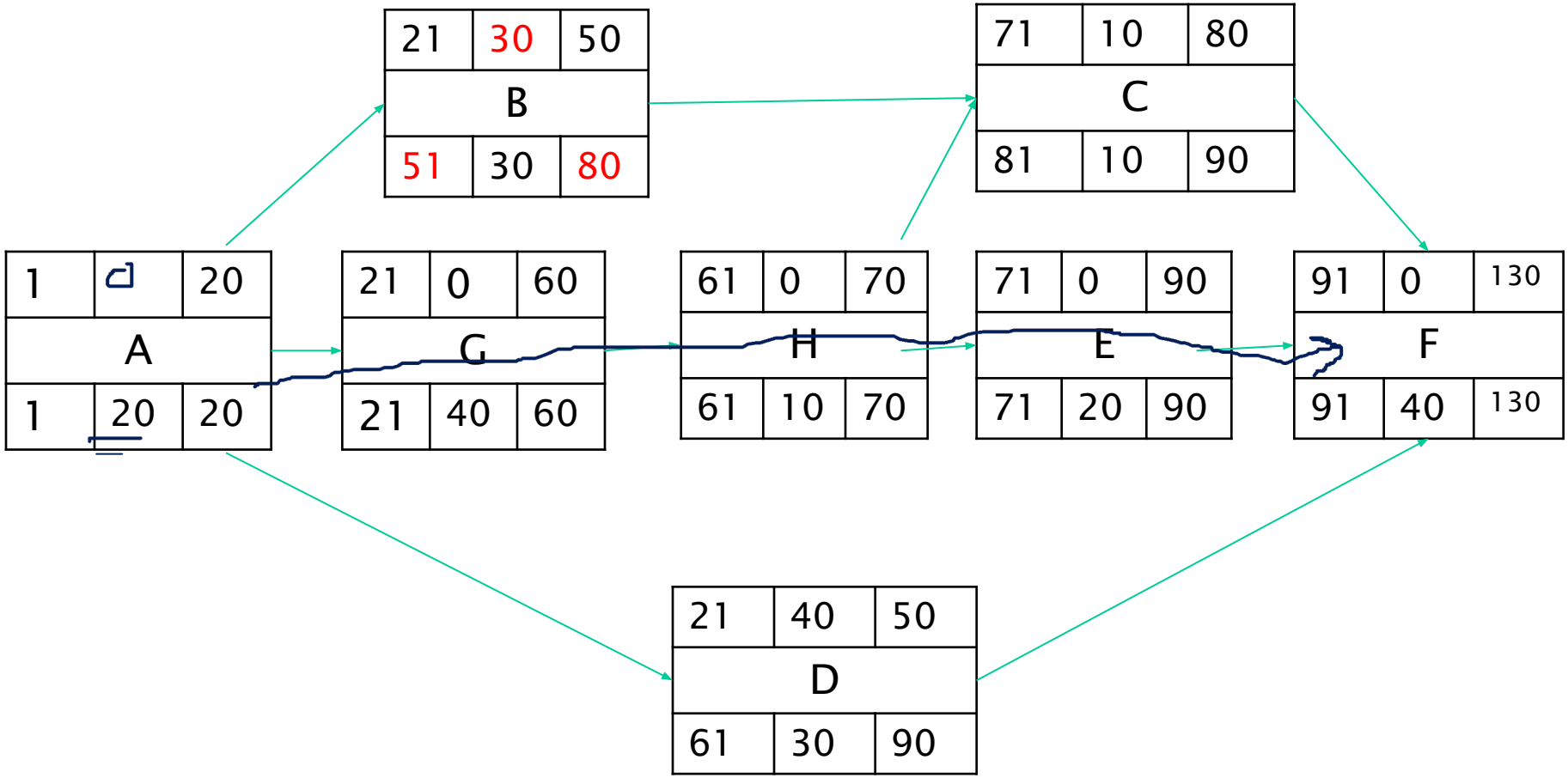
  = latest start time – earliest start time

Exercise

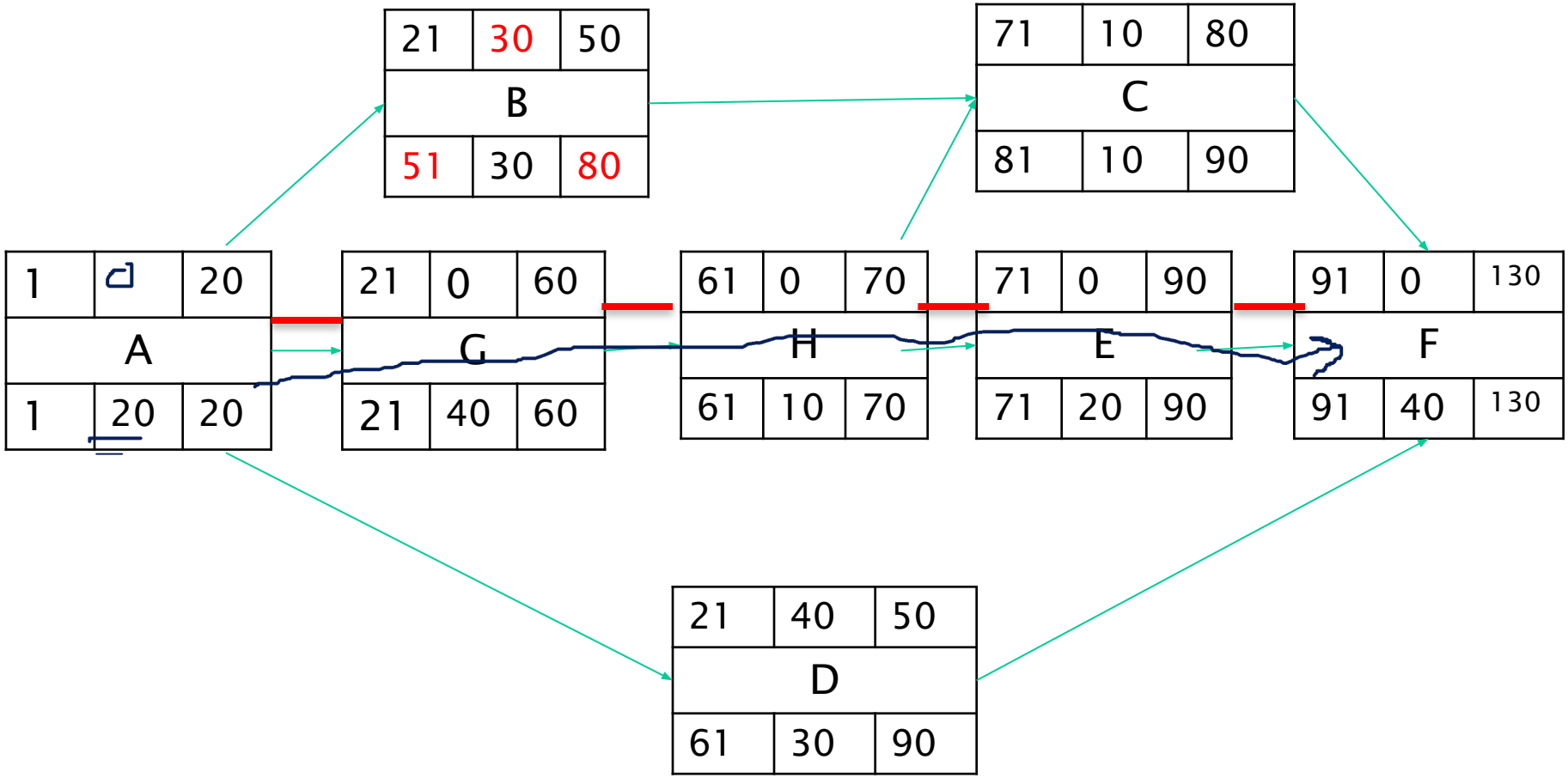
EST	Slack	EFT
A		
LST	20	LFT



Exercise


21 30 50
B
51 30 80
71 10 80
C
81 10 90
1 20
A
1 20 20
21 0 60
G
21 40 60
61 0 70
H
61 10 70
71 0 90
E
71 20 90
91 0 130
F
91 40 130
21 40 50
D
61 30 90

Exercise


21 30 50
B
51 30 80
71 10 80
C
81 10 90
1 20
A
1 20 20
21 0 60
G
21 40 60
61 0 70
H
61 10 70
71 0 90
E
71 20 90
91 0 130
F
91 40 130
21 40 50
D
61 30 90

Project Table

Work tasks	Planned start	Actual start	Planned complete	Actual complete	Assigned person	Effort allocated	Notes
I.1.1 Identify needs and benefits							Scoping will require more effort/time
Meet with customers	wk1, d1	wk1, d1	wk1, d2	wk1, d2	BLS	2 p-d	
Identify needs and project constraints	wk1, d2	wk1, d2	wk1, d2	wk1, d2	JPP	1 p-d	
Establish product statement	wk1, d3	wk1, d3	wk1, d3	wk1, d3	BLS/	1 p-d	
Milestone: Product statement defined	wk1, d3	wk1, d3	wk1, d3	wk1, d3			
I.1.2 Define desired output/control/input (OCI)							
Scope keyboard functions	wk1, d4	wk1, d4	wk2, d2		BLS	1.5 p-d	
Scope voice input functions	wk1, d3	wk1, d3	wk2, d2		JPP	2 p-d	
Scope modes of interaction	wk2, d1		wk2, d3		MLL	1 p-d	
Scope documents diagnosis	wk2, d1		wk2, d2		BLS	1.5 pd	
Scope other WP functions	wk1, d4	wk1, d4	wk2, d3		JPP	2 p-d	
Document OCI	wk2, d1		wk2, d3		MLL	3 p-d	
FTR: Review OCI with customer	wk2, d3		wk2, d3		all	3 p-d	
Revise OCI as required	wk2, d4		wk2, d4		all	3 p-d	
Milestone: OCI defined	wk2, d5		wk2, d5				
I.1.3 Define the Function/behavior							

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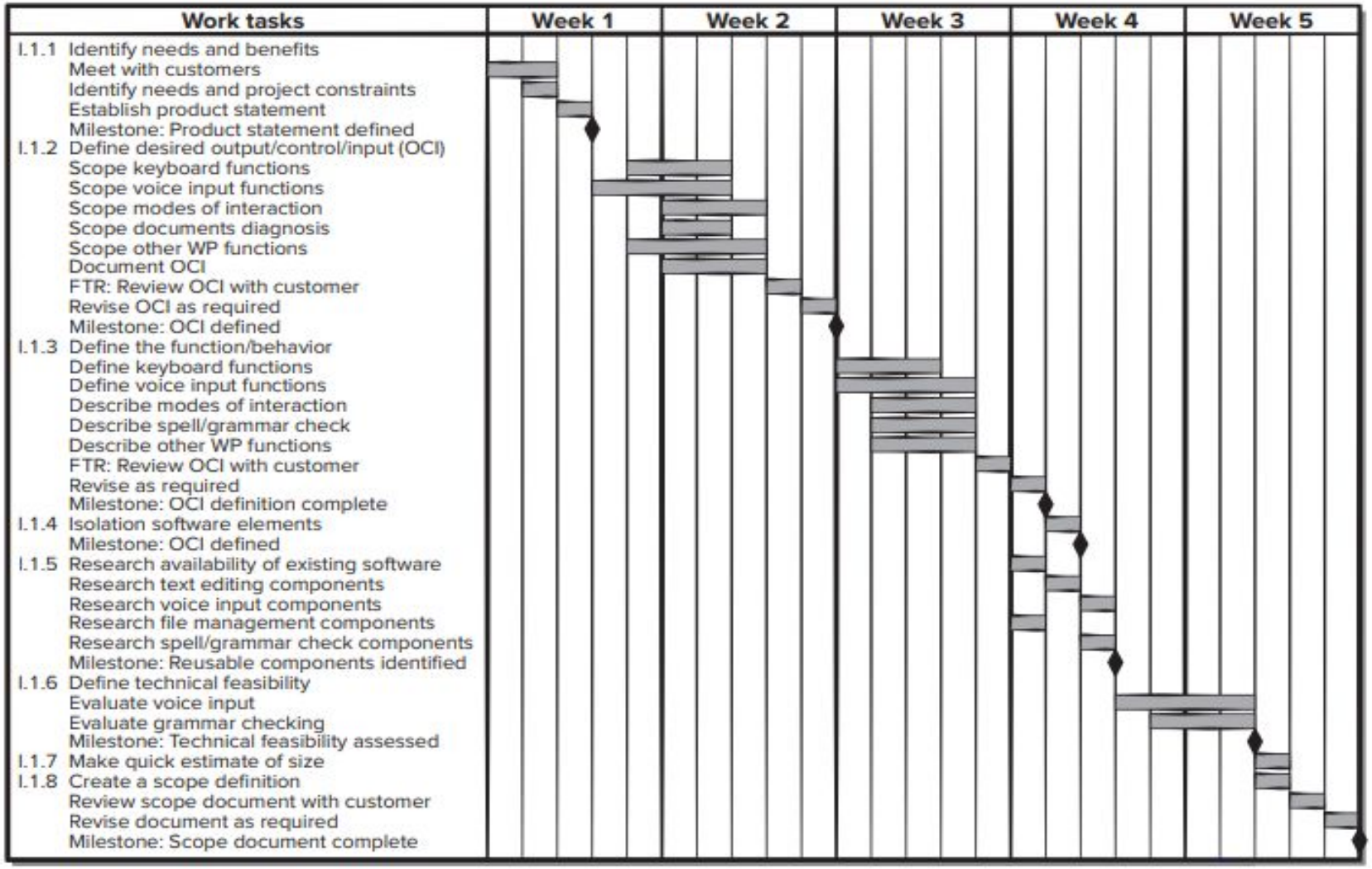
An Example Gantt (Timeline) Chart



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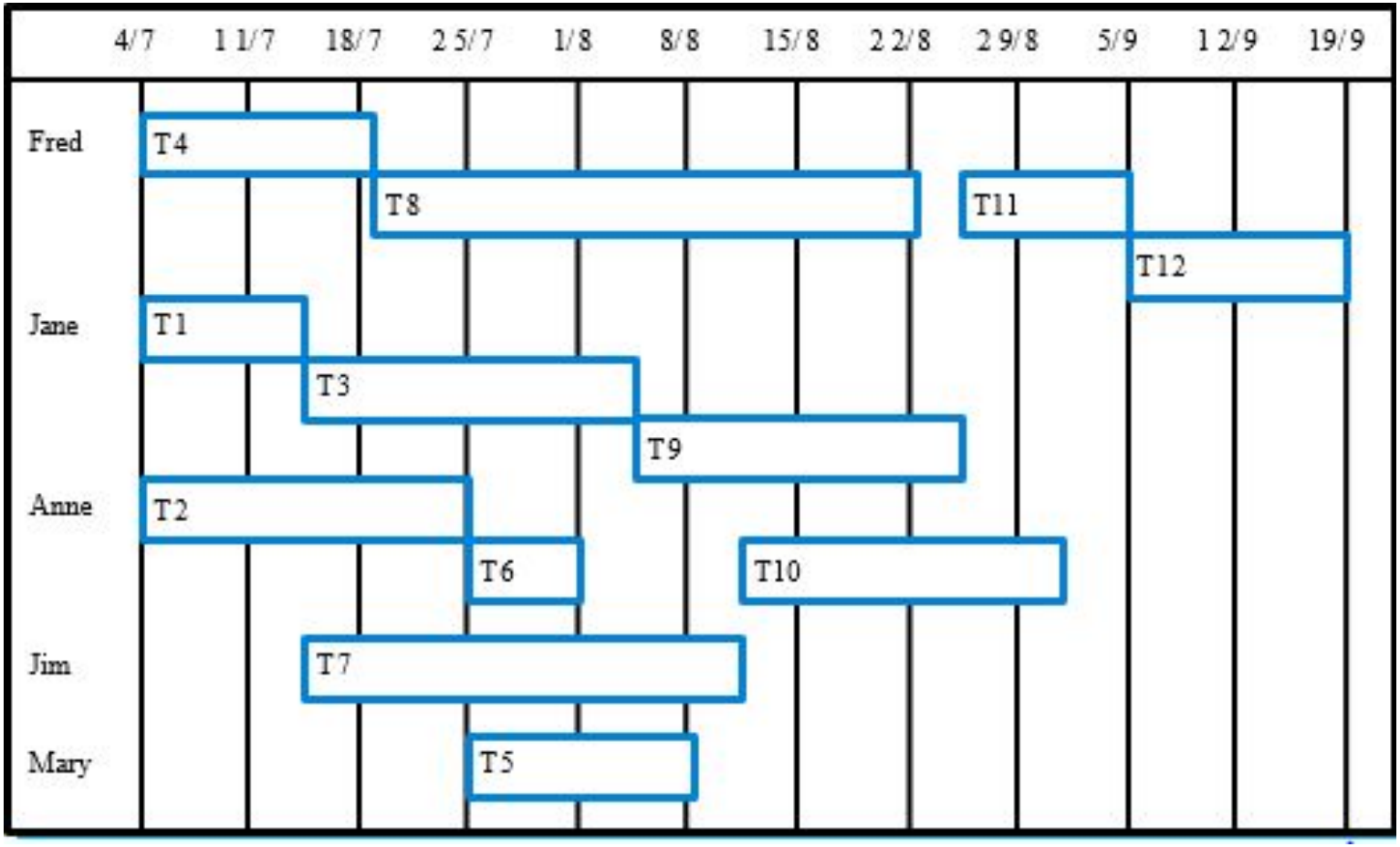
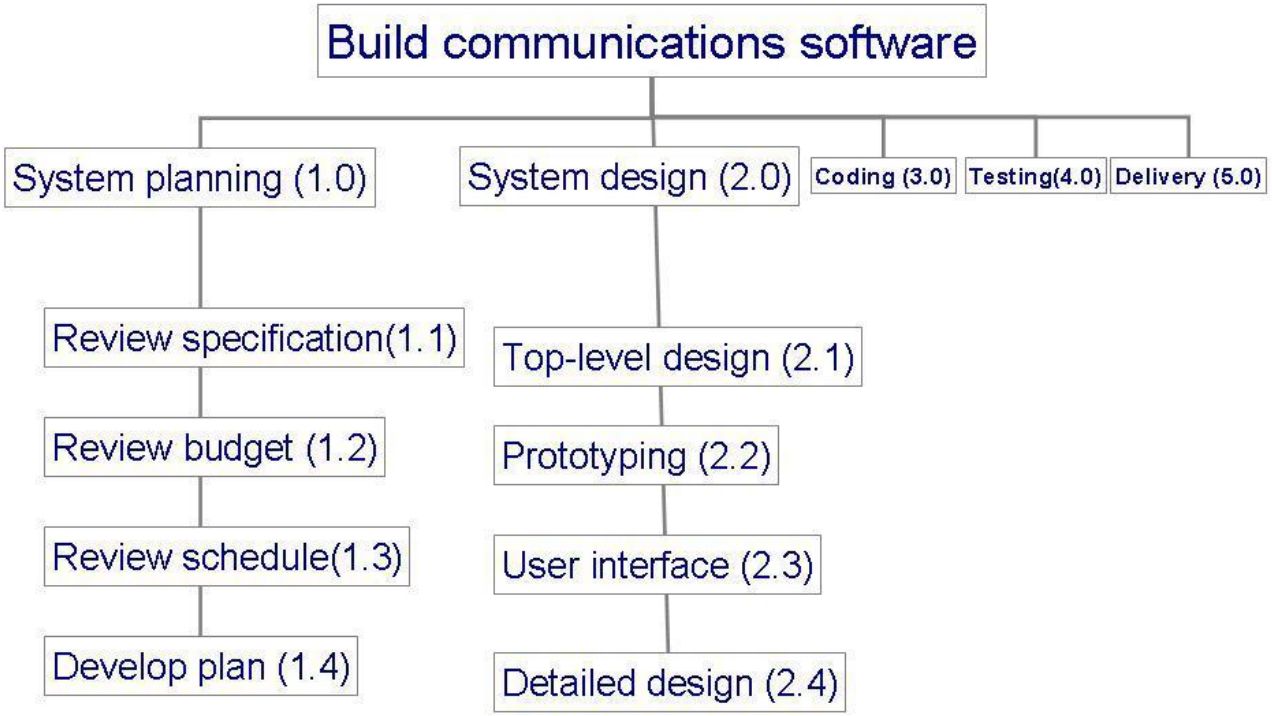
Gantt Chart


4/7
11/7
18/7
25/7
1/8
8/8
15/8
22/8
29/8
5/9
12/9
19/9
Fred
T4
T8
T11
T12
Jane
T1
T3
T9
Anne
T2
T6
T10
Jim
T7
Mary
T5

3.1 Tracking Progress

Tools to Track Progress

- Example: to track progress of building a communication software